UNIT 3

Exercise 2 : Semantic search using cosine similarity

Aim

To generate numerical embeddings for text and identify semantically similar sentences using embedding similarity.

Algorithm

- Start the program.
- Load a pretrained sentence-embedding model.
- Define a collection of text documents.
- Convert the documents into numerical vectors.
- Convert the user's query into an embedding.
- Calculate similarity scores between the query and documents.
- Display the most similar document.
- Stop.

Python Code and Output

```
query = "How do computers learn from information?"

query_embedding = model.encode(
    [query], normalize_embeddings=True
)

scores = embeddings @ query_embedding[0]

best_index = int(np.argmax(scores))

print("Query:", query)
print("Most similar document:", documents[best_index])
print("Similarity score:", round(float(scores[best_index]), 4))
```

```
Query: How do computers learn from information?
Most similar document: Machine learning allows computers to learn patterns from data.
Similarity score: 0.6238
```

Result

Text embeddings were generated successfully, and semantic similarity search identified the most relevant document.